
Which drink should we remove from the menu?
Brown Sugar Milk Tea $5.00
Classic Milk Tea $4.50
Matcha Latte $5.50
Taro Smoothie $6.00
Mango Fruit Tea $5.50

What problem are we solving?

Root problem: the lunch line at Mina's bubble tea shop is too long.

ROOT PROBLEM

The lunch line is too long

- Many students arrive at lunch at the same time.
- Staff must prepare five different drinks, so each order takes longer.
- Customers wait too long for their drinks.

MINA'S IDEA

Remove one drink from the menu

- Fewer drinks means faster preparation and a shorter line.
- But the shop must not lose much profit.
- So which drink should go?

WHAT WENT WRONG

AI chose the wrong drink

- Mina asked ChatGPT: "Which drink should I remove?"
- ChatGPT: "Remove Mango Fruit Tea, because it sells the fewest cups."
- But Mango Fruit Tea earns the **most** money per cup. The question did not tell the AI what matters.

Today's goal and a three-step method

Learn how to give business data to an AI tool so it reads the data correctly.

STEP 1

Explain the data

- What each column means.
- What unit it uses.
- How to calculate the measure you care about.

STEP 2

Get a table, then check one row

- Ask the AI to show results in a table.
- Check one row yourself with a calculator.

STEP 3

Ask for a decision

- State your goal.
- Ask for facts and guesses separately.
- Then ask for the recommendation.

One week of sales at Mina's shop

Fictional data created for this class. File: data/session_2_1_bubble_tea_sales.csv

drink	cups_sold	price_per_cup_USD	ingredient_cost_per_cup_USD
Brown Sugar Milk Tea	420	5.00	2.40
Classic Milk Tea	380	4.50	1.60
Matcha Latte	160	5.50	2.10
Taro Smoothie	110	6.00	4.80
Mango Fruit Tea	80	5.50	1.30

USD means United States dollars. Every value is per cup, except cups_sold, which is the total for the week.

Price, ingredient cost, and revenue

Before you ask an AI to analyze sales, know what the numbers mean. Example: one cup of Brown Sugar Milk Tea.

WORD 1

Price

What one customer pays for one cup.

$5.00

WORD 2

Ingredient cost

What the shop pays for the tea, milk, sugar, and cup used to make one cup.

$2.40

WORD 3

Revenue

All the money customers paid.

```
Revenue = cups sold x price
```

```
420 x $5.00 = $2,100.00
```

Revenue is **not** the money the shop keeps. Ingredients still have to be paid for.

Gross profit: the money left after ingredients

This is the measure Mina needs. It is the key idea of this session.

WORD 4

Gross profit

```
Gross profit per cup = price - ingredient cost
```

```
$5.00 - $2.40 = $2.60 per cup
```

```
Weekly gross profit = cups sold x gross profit per cup
```

```
420 x $2.60 = $1,092.00
```

BE CAREFUL

Gross profit is not final profit

- Gross profit subtracts **only** ingredient cost.
- It does not subtract rent or staff wages.
- So it is not the same as the money Mina takes home.
- This data can support gross profit, and nothing more.

"Sells the most" and "earns the most" are different

Compare two drinks from the table.

Drink	Cups sold	Gross profit per cup	Weekly gross profit
Mango Fruit Tea	80 (fewest)	$4.20 (highest)	$336.00
Taro Smoothie	110	$1.20 (lowest)	$132.00 (lowest)

READ THE TABLE

What it shows

- Mango sells the fewest cups, but each cup earns $4.20.
- Taro sells more cups, but its ingredients are expensive. Each cup earns only $1.20.

THE DECISION

What Mina loses each week

- Remove Mango: lose **$336.00** of gross profit.
- Remove Taro: lose only **$132.00**.

How the three-step method fixes the question

When a question gives no rule for deciding, AI often uses the easiest number to see. The question was incomplete.

STEP 1 FIXES

The definition problem

- The AI now knows that "profit" means gross profit.
- It knows how to calculate it.

STEP 2 FIXES

The trust problem

- One shown calculation lets you check the math in one minute.
- AI tools can make arithmetic mistakes. Always check at least one row.

STEP 3 FIXES

The decision problem

- The AI knows the goal: shorter line, least profit lost.
- It must say what it is guessing, such as how long each drink takes to make.

Poor prompt 1

Same bubble tea data. Compare what the prompt asks for.

PROMPT TO AVOID

The prompt

```
Here is my bubble tea shop's sales data for one
week.
Which drink should I remove from the menu?

drink,cups_sold,price_per_cup_USD,ingredient_cost_pe
r_cup_USD
Brown Sugar Milk Tea,420,5.00,2.40
Classic Milk Tea,380,4.50,1.60
Matcha Latte,160,5.50,2.10
Taro Smoothie,110,6.00,4.80
Mango Fruit Tea,80,5.50,1.30
```

WHAT THE AI OFTEN ANSWERS

A typical result

"Remove Mango Fruit Tea because it has the lowest sales."

WHY IT IS WEAK

Reasons

- It gives no rule for deciding, so the AI uses the most visible number: cups sold.
- It does not state the goal: a shorter lunch line while keeping as much profit as possible.
- The result is a confident answer that removes the drink with the highest profit per cup.

Poor prompt 2

Same bubble tea data. Compare what the prompt asks for.

PROMPT TO AVOID

The prompt

```
Calculate the profit of each drink and give me the
exact answer.

drink,cups_sold,price_per_cup_USD,ingredient_cost_pe
r_cup_USD
Brown Sugar Milk Tea,420,5.00,2.40
Classic Milk Tea,380,4.50,1.60
Matcha Latte,160,5.50,2.10
Taro Smoothie,110,6.00,4.80
Mango Fruit Tea,80,5.50,1.30
```

WHAT THE AI OFTEN ANSWERS

A typical result

Some answers call revenue "profit." Other answers invent rent or wage costs that are not in the data.

WHY IT IS WEAK

Reasons

- "Profit" has several meanings. This data can only support gross profit, and the prompt does not say so.
- Asking for an "exact answer" does not make the answer correct. It only makes the AI sound more certain.
- The prompt does not ask the AI to show its calculation, so you cannot check it.

Improved prompt 1

Steps 1 and 2. Copy this prompt exactly.

```
I run a small bubble tea shop. Below is one week of
sales data.

What each column means:
- cups_sold: number of cups sold this week
- price_per_cup_USD: what one customer pays for one
cup, in US dollars
- ingredient_cost_per_cup_USD: cost of the ingredients
for one cup, in US dollars

Use these definitions:
- Revenue = cups_sold x price_per_cup_USD
- Gross profit per cup = price_per_cup_USD -
ingredient_cost_per_cup_USD
- Weekly gross profit = cups_sold x gross profit per
cup
Gross profit does not include rent or staff wages.
```

```
Make a table with these columns: drink, cups_sold,
revenue,
gross profit per cup, weekly gross profit.
Sort the table by weekly gross profit, from highest to
lowest.
Show the calculation for the Taro Smoothie row so I
can check it.
Do not recommend anything yet.

drink,cups_sold,price_per_cup_USD,ingredient_cost_per_
cup_USD
Brown Sugar Milk Tea,420,5.00,2.40
Classic Milk Tea,380,4.50,1.60
Matcha Latte,160,5.50,2.10
Taro Smoothie,110,6.00,4.80
Mango Fruit Tea,80,5.50,1.30
```

Why improved prompt 1 works

Each part of the prompt solves one problem.

WHY IT WORKS

What each part does

- It explains each column and its unit, so the AI cannot misread the data.
- It gives the formula for gross profit, which removes the confusion in Poor Prompt 2.
- It asks for a sorted table, so the five drinks are easy to compare.
- It asks to see one calculation, so you can check the math.
- "Do not recommend anything yet" keeps calculating and deciding as separate steps.

Improved prompt 2

Step 3, sent in the same chat after Improved Prompt 1. Copy this prompt exactly.

```
I checked your table with a calculator. Weekly gross
profit is:
Brown Sugar Milk Tea $1,092.00, Classic Milk Tea
$1,102.00,
Matcha Latte $544.00, Taro Smoothie $132.00, Mango
Fruit Tea $336.00.

My goal: remove one drink to make the lunch line
shorter,
while losing as little weekly gross profit as
possible.
```

```
Answer in three parts:
1. Facts: what the table shows. Use only the numbers
above.
2. Guesses: what you are assuming that the data does
not show.
3. Recommendation: which drink to remove, in one
sentence, and why.
Then explain in two sentences why "remove the drink
with the lowest
sales" is a weak rule for this shop.
```

Why improved prompt 2 works

Each part of the prompt solves one problem.

WHY IT WORKS

What each part does

- It gives the checked numbers, so the AI does not recalculate and make a new mistake.
- It states the goal and the decision rule: shorter line, least gross profit lost.
- Separating facts from guesses stops the AI from presenting unknown details, such as preparation time, as facts.
- The last request makes the AI explain the mistake in Poor Prompt 1, which checks the main lesson.

Exercise 1: Compare a poor prompt with an improved prompt

Guided practice. Follow the five steps in order.

GUIDED PRACTICE

Steps

1. Open ChatGPT and start a **new chat**. Paste and send **Poor Prompt 1**. Write down the drink the AI chose and its reason.
2. Start another **new chat**, so the first answer does not affect the second. Paste and send **Improved Prompt 1**.
3. Check the Taro Smoothie row with a calculator: $6.00 - $4.80 = $1.20 per cup, and 110 x $1.20 = $132.00. Compare the AI table with the answer key. Circle any wrong row.
4. In the same chat, paste and send **Improved Prompt 2**.
5. For the Results section of your report, write the drink chosen by Poor Prompt 1, the drink chosen by Improved Prompt 2, and one sentence that explains why they are different.

Answer key: weekly gross profit

Use this table to check the AI. Sorted from highest to lowest weekly gross profit.

Drink	Revenue	Gross profit per cup	Weekly gross profit
Classic Milk Tea	$1,710.00	$2.90	$1,102.00
Brown Sugar Milk Tea	$2,100.00	$2.60	$1,092.00
Matcha Latte	$880.00	$3.40	$544.00
Mango Fruit Tea	$440.00	$4.20	$336.00
Taro Smoothie	**$660.00**	**$1.20**	**$132.00**

Notice: Brown Sugar Milk Tea has the highest revenue, but Classic Milk Tea has the highest weekly gross profit.

Exercise 2: One number changes

The price of milk goes up. Does Mina's decision still hold?

THE CHANGE

Milk costs more

- Four drinks use milk: Brown Sugar Milk Tea, Classic Milk Tea, Matcha Latte, and Taro Smoothie.
- Their ingredient cost rises by **$0.50 per cup**.
- Mango Fruit Tea has no milk. Its cost does not change.

PROMPT TEMPLATE

Replace each part in [square brackets]

```
Continue with the same sales data and the same
definitions.
One thing has changed: [what changed, which
drinks, and by how much].
Recalculate [which values] and show the new table
sorted by [which column].
Show the calculation for the [drink name] row so I
can check it.
Then answer: should I still remove [drink name]?
Separate facts and guesses, and give a one-
sentence recommendation.
```

Part A: Worked example (run it)

In the chat from Exercise 1, copy and send this completed prompt.

COMPLETED TEMPLATE

The prompt

```
Continue with the same sales data and the same
definitions.
One thing has changed: the ingredient cost of
Brown Sugar Milk Tea,
Classic Milk Tea, Matcha Latte, and Taro Smoothie
rose by $0.50 per cup.
Mango Fruit Tea did not change.
Recalculate gross profit per cup and weekly gross
profit, and show the
new table sorted by weekly gross profit, from
highest to lowest.
Show the calculation for the Taro Smoothie row so
I can check it.
Then answer: should I still remove Taro Smoothie?
Separate facts and guesses, and give a one-
sentence recommendation.
```

CHECK THE ANSWER

Taro Smoothie row

```
$6.00 - $5.30 = $0.70 per cup
```

```
110 x $0.70 = $77.00
```

- Taro Smoothie is still the lowest, so the decision holds.
- Total weekly gross profit falls from $3,206.00 to $2,671.00, a drop of $535.00.

Part B: Your turn (write it)

Mina's friend says: "Do not remove Taro Smoothie. Raise its price from $6.00 to $7.00 instead."

YOUR TASK

Fill in the template yourself

- Keep the milk cost increase from Part A.
- Your prompt must name: the change, the drink, the values to recalculate, and the row to check.
- Send it in the same chat.

CHECK THE ANSWER

Taro Smoothie at $7.00

```
$7.00 - $5.30 = $1.70 per cup
```

```
110 x $1.70 = $187.00
```

- Still the lowest of the five drinks.
- A good AI answer lists a **guess**: the data does not show whether 110 customers will still buy Taro Smoothie at the higher price.

Write your session report

Use Report_template.docx in Microsoft Word. Save it as FirstName_StudentID_W5_Keyword.docx and upload it to iLearn by 23:59 on class day.

SUBMIT

One report for this session

- **Introduction:** Explain Mina's root problem (the long lunch line), her idea to remove one drink, and why ChatGPT's first answer was wrong.
- **Method:** Copy Poor Prompt 1, Improved Prompts 1 and 2, the Part A prompt, and the Part B prompt you wrote.
- **Results:** Include the drink chosen by Poor Prompt 1 and by Improved Prompt 2, with one sentence that explains the difference. Add the AI gross profit tables and your hand check of the Taro Smoothie row for Exercise 1 and Part A.
- **Conclusion:** Give your one-sentence recommendation for Part B. Name one fact from the data that supports it and one guess the data cannot confirm.

How to know the work is complete

Check your report before you hand it in.

VERIFY

Success check

- You can explain in one sentence why "remove the drink with the lowest sales" was the wrong rule for Mina's shop.
- Your Part B prompt produces a table you checked with a calculator.